

Cauldron fills with powder snow in frozen ocean biome while it's visually raining

MC-230678 Particles

Status	Resolved / Fixed
Affected	1.17, 1.17.1 Pre-release 2, 1.17.1, 21w37a, 1.18 Pre-release 6, 1.18 Pre-release 7 (+12 more)
Fixed in	23w03a
Reported	2021-06-29

STEPS TO REPRODUCE

1. Launch Minecraft and enter a Creative world.
2. Find a frozen ocean biome with open sky above the area.
3. Use `/fill ~10 ~ ~10 ~ ~10 minecraft:cauldron`` to create a large layer of cauldrons.
4. Set the weather to rain with `/weather rain``.
5. Wait at least a few minutes for the cauldrons to be affected by the weather.
6. Observe that some cauldrons contain powder snow even in a local region where rain is visibly falling.

OBSERVED BEHAVIOUR

Actual behavior

In a **frozen ocean** biome, exposed cauldrons are filled with **powder snow** even in areas where the weather is rendered as rain. The cauldron's contents therefore do not match the precipitation displayed at its position.

This occurs across the biome's local temperature gradient: ice formation and snow layers respect the local temperature values, but cauldrons consistently collect powder snow instead of collecting water in the regions where rain is visually falling.

The issue is not reproducible in **deep frozen ocean** biomes.

EXPECTED BEHAVIOUR

In a frozen ocean, cauldrons exposed to the sky should fill according to the local weather precipitation: water where the area is visually raining and powder snow where it is snowing. Their contents should follow the same local temperature boundaries that determine ice and snow formation, rather than filling with powder snow throughout the biome.

ENVIRONMENT

Minecraft Java Edition Versions: 1.17, 1.17.1 Pre-release 2, 1.17.1, 21w37a, 1.18 Pre-release 6, 1.18 Pre-release 7, 1.18.1, 22w11a, 22w14a, 22w18a, 1.19, 1.19.1 Pre-release 4, 1.19.1 Pre-release 5, 1.19.1, 1.19.2, 22w42a, 1.19.3 Pre-release 2, 1.19.3 (includes snapshots)

ORIGINAL DESCRIPTION

The bug

Cauldrons are supposed to be filled with water while it's raining, and powder snow while it is snowing. Frozen oceans have a unique temperature gradient that makes it snow in certain places and rain in others. However, whenever a cauldron is placed in this biome, it is filled with powder snow regardless of which local region it is in, even if it is visually raining. Ice and snow are formed as normal (i.e. following the boundaries of the local temperature values).

How to reproduce

- Find a frozen ocean, or create a superflat world with just frozen ocean
- Use `/fill` command to create a large layer of cauldrons under open sky
- Change weather to raining/snowing
- Observe that some cauldrons are filled with powder snow even if it is raining
- Replace the platform with solid blocks, and see how snow layers are formed normally

This bug does not exist for the deep frozen ocean biome.

Code analysis

Code analysis can be found in this comment.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-230678>